

商の微分の証明

関数 $f(x), g(x)$ が微分可能であるとき

$$1. \left(\frac{f(x)}{g(x)} \right)' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$2. \left(\frac{1}{g(x)} \right)' = -\frac{g'(x)}{(g(x))^2}$$

(2の証明)

$$\begin{aligned} \left(\frac{1}{g(x)} \right)' &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1}{g(x+h)} - \frac{1}{g(x)} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{g(x) - g(x+h)}{g(x+h) \cdot g(x)} \\ &= \lim_{h \rightarrow 0} \left(-\frac{g(x+h) - g(x)}{h} \cdot \frac{1}{g(x+h)g(x)} \right) \\ &= -\frac{g'(x)}{(g(x))^2} \quad \square \end{aligned}$$

(1の証明)

$$\begin{aligned} \left(\frac{f(x)}{g(x)} \right)' &= \left(f(x) \cdot \frac{1}{g(x)} \right)' \\ &= f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \left(\frac{1}{g(x)} \right)' \quad \leftarrow \text{積の微分} \\ &= \frac{f'(x)}{g(x)} + f(x) \cdot \frac{-g'(x)}{(g(x))^2} \\ &= \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \quad \square \end{aligned}$$